

AE-343: Synthesis of [Ru(CO)(OH)₂(PNN)] using Ag₂O, 5 h

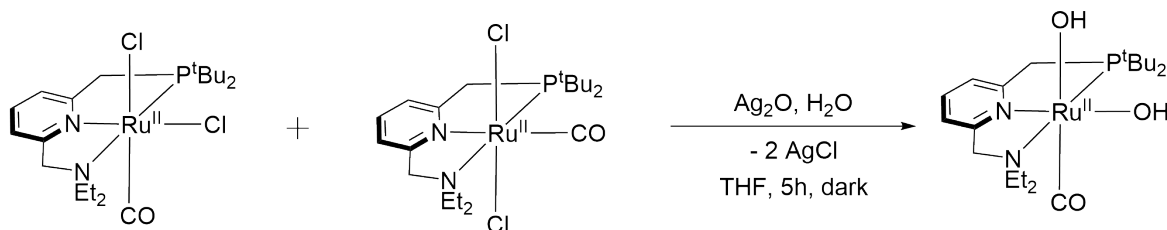
Date: 2024-09-10

Tags: PNN [Ru(CO)(OH)₂(PNN)]
Dihydroxo Ag₂O NMR AE 1H 31P

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure



[RuCl₂(CO)(PNN)]

Molecular Weight:
522.46 g/mol

Silver Oxide

Molecular Weight:
231.74 g/mol

[Ru(CO)(OH)₂(PNN)]

Molecular Weight:
485.57 g/mol

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	AEI-010: Synthesis of [Ru(CO)(OH) ₂ (PNN)] using Ag ₂ O Experiment - AE-252: Synthesis of [Ru(CO)(OH) ₂ (PNPtBu)] using Ag ₂ O, 5 h AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNPtBu)] using Ag ₂ O, 5 h

Reagents

Name	CAS-Number/Experiment-Number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume _{theo} [ml]	Volume _{exp} [ml]	Density [g/mL]
[RuCl ₂ (CO)(PNN)]	AE-340: Synthesis of [RuCl ₂ (CO)(PNN)] via [RuCl ₂ (CO)(p-cymene)]	0.231	1.00	/	120.82	522.46	/	/	/
Ag ₂ O	20667-12-3	1.42	6.14	329.07	329.24	231.74	/	/	/
H ₂ O	Experiment - KRA-104: Degassing of Milli-Q water	54.11	234	975	/	18.02	0.973	0.97	0.998
THF	Experiment - KRA-080: Degassing of THF	/	/	/	/	/	6.9	7.0 + 7.5	/
Et ₂ O	JU-KRA-172: Degassing of Diethylether	/	/	/	/	/	/	8.5	/

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an nitrogen filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Due to the instability of Ag_2O and Dihydroxo to the light, the procedure was done mostly in the dark. The precise measures, which were taken, are listed in "**Steps**".

Date	Time	Step	Observations
10.09		A 25mL-brown glas Schlenk flask was prepared, by heating under vacuum	
	9:40	$[\text{RuCl}_2(\text{CO})(\text{PNN})]$ and Ag_2O was weighted in air into the reaction schlenk flask and the flask was set under argon.	
		The light in the fumehood was turned out	
	10:10 - 10:12	THF (7.0 mL) and water (0.97 mL) were added to the reaction schlenk flask.	
		The flask was wrapped in aluminium foil	
	10:12 - 15:12	The reaction mixture was stirred for 5 h at 1000 rpm.	
	15:13 - 15:23	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 50 mL-Schlenk	
	15:23 - 15:30	The reaction flask was washed with MeCN (2 * 1.0 mL)	Comined filtrates: AE-343-1 After filt.jpg
	15:40 - 17:35	The solvents of AE-343-1 was removed under reduced pressure.	To avoid bubbling up of the mixture the water bath for warming was removed when approx. 2 mL solvents were left and the mixture was allowed to freeze --> worked well, but drying takes significant longer
		The obtained solid was stored in the dark at -20 °C	

12.09		The solid was allowed to warm to room temperature	solid.jpg
	9:51	To the solid THF (7.5 mL) was added und the obtained mixture was shaken for approx. 30 s and stirred	in THF.jpg
	10:17	Et2O (8.5 mL) was added to the mixture	with Et2O.jpg
	10:23	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 50 mL-Schlenk	
	10:35 - 10:40	The solid was washed with Et2O (3 * 1 mL)	
	10:42 - 13:30	The solid was dried under reduced pressure	
	13:55	The product was transfered into a vial in a Schlenk flask	
	14:15	An NMR sample was prepared from AE-343-1 (approx. 5 mg) in DMSO-d6 according to Protocol - Preparation of NMR Sample	
		The remaining solid was stored in the dark under argon at -20 °C	

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
12.09	17:50	AE-343-1	NMR 1H, 31P{1H}quant, 1Hzgespg	IAAC 400 MHz I	DMSO-d6	AE-343-1.zip	AE-343-1_10.nmrium	https://doi.org/10.1039/D1EE01053K	Main signal is desired product with 90 %. Three impurites are seen (all small, but distinct signals, but integration not reliable) at: 147, 109 and 57 ppm

Product characterization and results

Sample	Mass [mg]	Purity	Mass _{pure} [mg]	Amount [μmol]	Yield [%]	Description	Foto	Storage location
AE-343-1	35.67	0.90	32.10	66.1	28	grayish sligthly greenish solid		Equipment - Freezer Lab E004 CEEC I Level A

Linked experiments

- AEI-010: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNN})]$ using Ag_2O
- JSC-KS-04: Degassing of n-Heptane
- AEI-024: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{NNN})]$ with 2,6-bis(N,N-diethylaminomethyl) using Ag_2O
- AEI-026: Capillary production
- AEI-048: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{NNN})]$ with 2,6-bis(N,N-diisopropylaminomethyl) using Ag_2O
- AEI-064: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AEI-073: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-101: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-102: Degassing of THF
- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-108: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-114: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-121: Degassing of n-Heptane
- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-157: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-167: Irradiation of PhPDA (365 nm + white LED, gas phase measurement), AE-151, 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-198: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ with Ag_2O with different reaction times (1, 2, 5, 24 h)
- AE-KRA-204: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h
- AE-208: Degassing of Milli-Q water
- KRA-080: Degassing of THF
- KRA-104: Degassing of Milli-Q water

- AE-230: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-232: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h
- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white
- AE-252: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h
- AE-261: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-281: Stability test of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-252) under air and afterwards irradiation with 365 nm and white
- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h
- JU-KRA-172: Degassing of Diethylether
- AE-340: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNN})]$ via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

Linked resources

- Equipment - [Freezer Lab E004 CEEC II](#)
- Protocol - [Preparation of NMR Sample](#)
- Protocol - [Filtration with frit technique](#)
- Protocol - [Schlenk Technique](#)
- Protocol - [Preparation of capillary for H₂O-NMR](#)

Attached files

AE-343-1_10.nmrium
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After-filt.jpg

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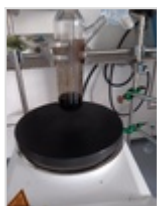
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AE-343.cdxml

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AE-343.png

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Unique eLabID: 20240910-d049ab9f1f46bfb369fe2c1e2fa9d9dd63e98cdc
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1343>